

# When is expensive perception worth paying for?

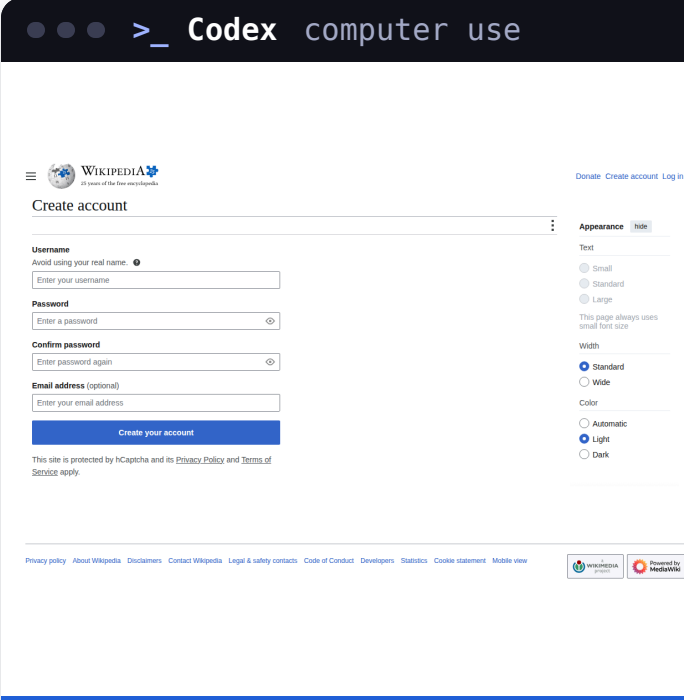
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MSc thesis · REALM workshop, EMNLP 2026

# A web agent works the page for you — and can see it three ways.

• • • > **Codex** computer use



**LOOK** · screenshot

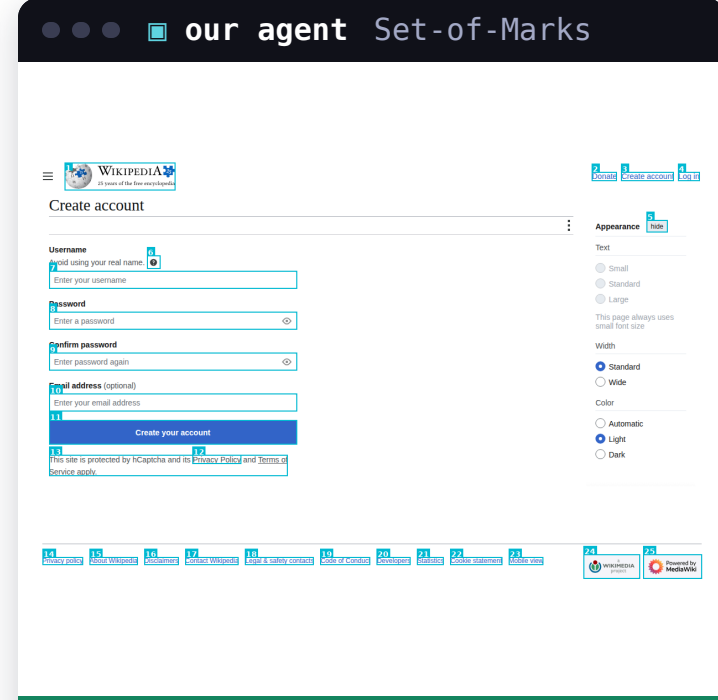
• • • \* **Claude Code** Playwright MCP

```
playwright · browser_snapshot – real reply 161 lines

- generic [active] [ref=e1]:
: 26 lines
- heading "Create account" [level=1] [ref=e35]
: 55 lines
- generic "Create account" [ref=e53]:
: 6 lines
  - button "username policy" [ref=e62]
  - textbox "Username" [ref=e66]:
    - /placeholder: Enter your username
: 3 lines
  - textbox "Password" [ref=e72]:
    - /placeholder: Enter a password
: 4 lines
  - textbox "Confirm password" [ref=e79]:
    - /placeholder: Enter password again
: 3 lines
  - textbox "Email address (optional)" [ref=e86]:
    - /placeholder: Enter your email address
  - button "Create your account" [ref=e89]
: 51 more lines
```

**READ** · page as text

• • • **our agent** Set-of-Marks



**BOTH** · screenshot + marks

One real page, 15 Sep 2026 · frames name the tools, not their limits · marks by our code

ONE RECORDED RUN PER VIEW · NOTHING ON THIS SCREEN IS LIVE

■ Same task, three ways of seeing it, three different bills.

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■ Choose the right view for every task: 27 → 43 of 100 solved.

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Upper bound · the demo's site, 224 tasks · other two sites +11 and +16 per 100, -14 and -27% · not Claude or GPT results

IF YOU BUILD A WEB AGENT

# Should it look, read, or both — and can it learn to choose?

1

How do the views behave?

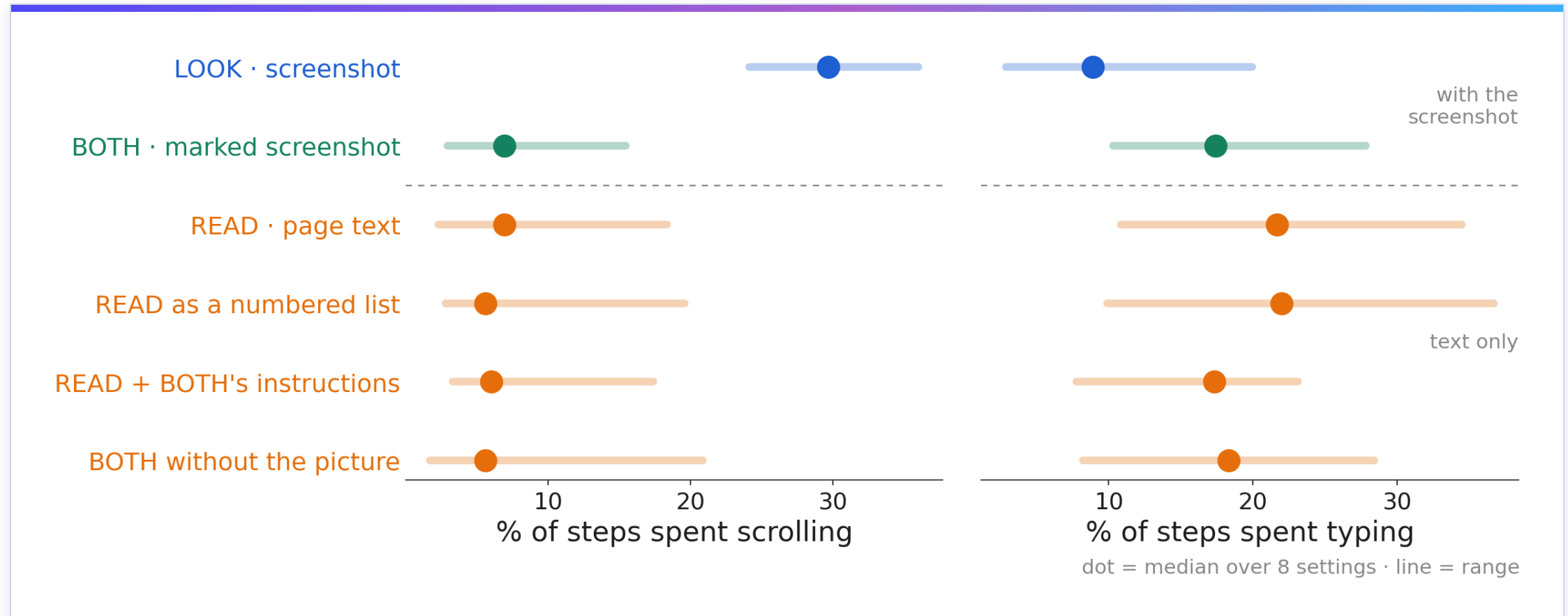
2

How do they fail?

3

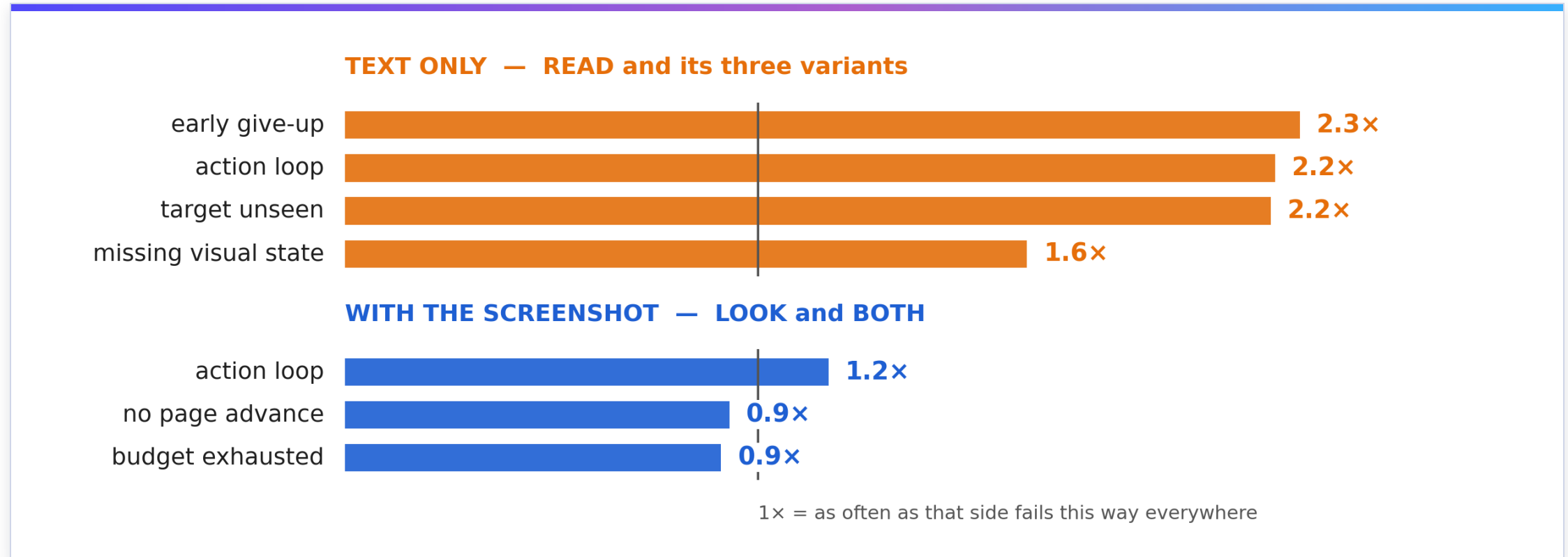
Can it be learned?

■ With only the screenshot, the agent scrolls far more and types far less.



Medians over 8 settings: LOOK scrolls in **about 30%** of its steps, every other view **about 6%**; it types about half as often

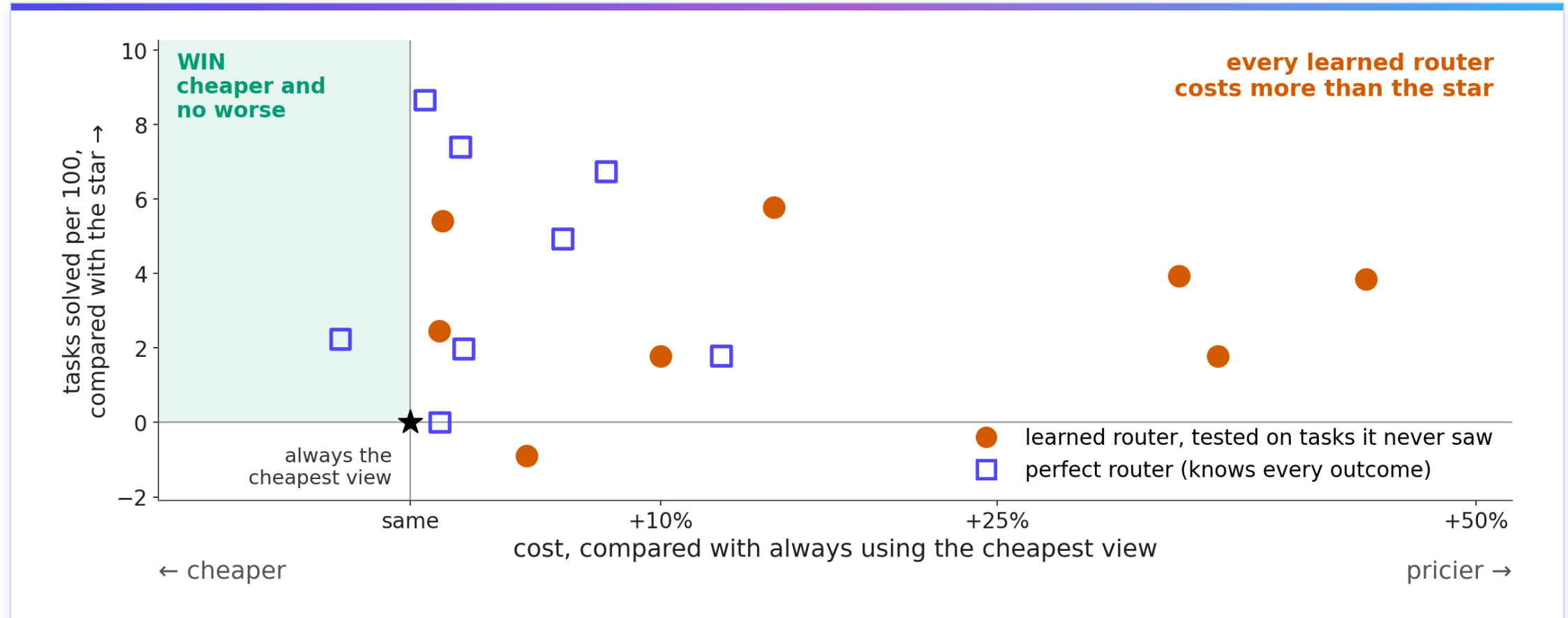
- Text-only views fail in ways you can name; views with the screenshot just never get there.



Each side vs its own failures elsewhere · text-only: several over **2x** · LOOK and BOTH: none stands out · six VisualWebArena settings

### ③ CAN A ROUTER LEARN WHICH VIEW TO USE?

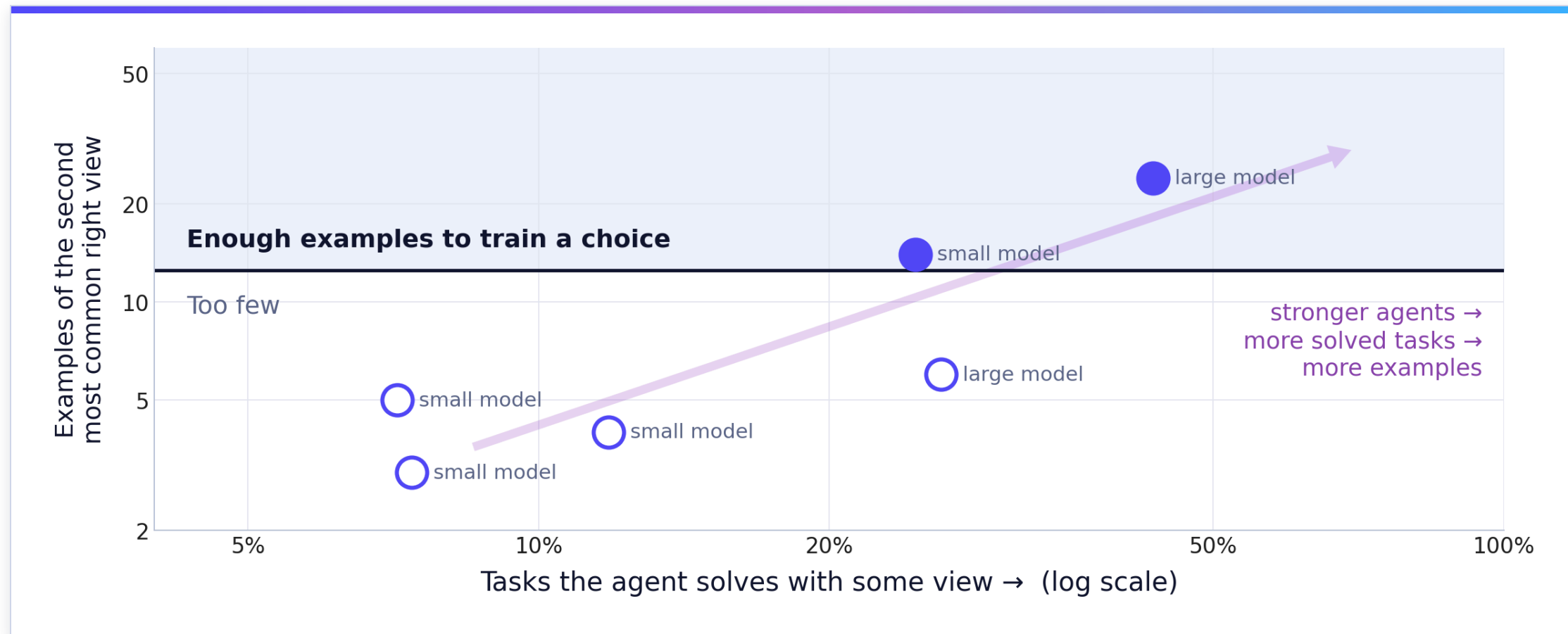
■ No learned router reaches the win corner — even a perfect one rarely does.



Learned routers: **0 of 8** settings

Perfect router: **1 of 8**

## ■ Routing's scaling law: no wins, no examples.



Six VisualWebArena settings · the arrow is a direction, not a fit · below the line, today's agents need at least **2–4×** more tasks

NOT YET

## ■ Improve the agent first, collect reliable examples, then learn when to look.

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### WHAT IT TAKES

- a stronger agent
- at least **2–4×** more tasks
- reruns that agree

### THREE ASKS

- ✓ type a task at the board
- ✓ which view do you use?
- ✓ lend us a stronger agent

Look, read, or both?  
It depends on the task.  
Nothing we trained knows it yet.



**Come to the board** — try your own task, live in all three views.

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REFERENCE – TASK 130, RECORDED STEPPING, 39 S